Clark Zhang

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Work Authorization: Canadian Citizen — Eligible for U.S. TN Status (No H-1B sponsorship required)

EDUCATION

University of British Columbia **Graduation: April 2026**

Bachelor of Applied Science in Integrated Engineering (Mechatronics Concentration) — cGPA: 4.0 (92%)

WORK EXPERIENCE

Mechatronics Engineering Project Lead, Hein Lab **December 2024 – December 2025**

- Led the end-to-end development of a robotic drug-analysis platform for Canada's overdose crisis response, automating high-precision sample preparation and chemical analysis for 1,500+ real samples with 99% overall reliability
- Reduced automation module costs by 90% by designing, fabricating, and validating custom robotic subsystems in SolidWorks and Fusion 360, including an end effector, vortexer, liquid handler, and BLDC centrifuge
- Produced 11 CAD assemblies and manufacturing drawing packages using GD&T principles for 200+ CNC-machined, waterjet-cut, and 3D-printed components in lab-grade robotic modules
- Developed a multi-tiered control architecture using Python to coordinate UR robot workflows and C++/ROS2 nodes to automate 12 hardware modules, monitor device states, and flag sensor-reported errors
- Built 2 custom PCBs in Altium to coordinate stepper motors, BLDC control, and sensor feedback, enabling 5,000+ automated sample-handling runs with repeatable module control

Mechatronics Engineering Intern, Hein Lab **May 2024 – December 2024**

- Commissioned 4 UR/KUKA industrial robot arms across 3 automation platforms, programming robot pathing, configuring I/O and networking, and validating reachability and collision clearance to reduce scientist manual sample-preparation time by ~80%
- Decreased robot workflow modification time by 75% by transitioning rigid PLC-based sequences into reusable Python control libraries for UR/KUKA arms
- Eliminated 90% of workflow errors before deployment by modeling robotic workcells in SolidWorks and simulating robot paths, reachability, collision clearance, and module access in RoboDK

Airfoils Team Lead, UBC AeroDesign **April 2022 – September 2025**

- Led an 11-person subteam designing and manufacturing a 120-inch heavy-lift RC aircraft for SAE AeroDesign, earning 2nd place overall in 2025 out of 40+ teams
- Resolved clearance and manufacturability issues before final aircraft assembly by designing the main wing, ailerons, servo mounts, and linkage interfaces in CATIA V5 and SolidWorks using DFMA and GD&T principles
- Improved lift performance by ~20% by validating aerodynamic and structural designs with SolidWorks FEA, Ansys Fluent, and wind-tunnel testing before manufacturing
- Reduced build errors by 70% by standardizing assembly and manufacturing processes for flight hardware made with 3D printing, CNC machining, waterjetting, laser cutting, and carbon-fiber prepreg layups

PROJECTS

Robotic Tool Vending Machine (Haven) **September 2025 – April 2026**

- Designed and validated a Cartesian XY gantry and rotary indexing storage mechanism for controlled checkout and return of 30 shared shop tools in SolidWorks, reducing tool loss to zero since deployment
- Developed Klipper-based motion control with magnetic encoder feedback and a custom I2C multiplexer PCB to coordinate gantry travel, storage indexing, and tool-position verification across 11 tool types
- Fabricated and iterated prototypes using 3D-printed parts, waterjet-cut plates, CNC-machined components, and manual shop tools

Autonomous Diver-Support Surface Vehicle for Debris Recovery **September 2024 – April 2025**

- Earned 1st place at the National CSME Design Competition by leading the development and manufacturing of an autonomous diver-support surface robot with a dual-pontoon flotation chassis, adjustable propulsion mounts, and mobile debris basket
- Improved saltwater durability and field readiness by building and assembling plywood platforms, titanium-fastened structures, aluminum brackets, and 3D-printed propulsion/basket mounts

SKILLS

Mechanical Design: SolidWorks, CATIA, Fusion 360, AutoCAD, SolidWorks FEA/Flow, Ansys FEA/Fluent, GD&T, DFMA

Robotics & Controls: Python, C++, ROS2, PLC (Siemens), UR/KUKA path planning, Git, PCB design (Altium), RoboDK

Manufacturing & Prototyping: CNC, waterjet/laser cutting, injection molding, 3D printing, composites, shop tools, drawings